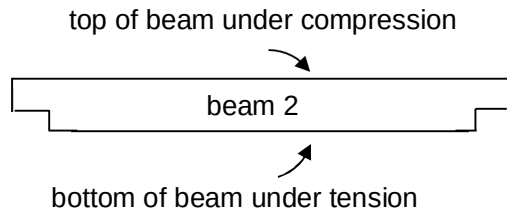


$$= 1.88 \times 10^4 \text{ eV}$$

A1

- 8 (a) It is the number of trains arriving at a station per unit time. A1
- (b) Aluminium alloy has high strength-to-weight ratio, thus reduces the amount of friction by reducing the weight of the trains. It has high corrosion resistance. Aluminium's natural passivation process in which a thin aluminium oxide layer forms when the metal is exposed to oxygen, reduces the possibility of further oxidation. A1

(c) (i)



Correct labelling of compression and tension

A1

- (ii) Total normal reaction forces = $(350 + 380) \times 10^3$
 $= 730 \times 10^3 \text{ N}$
- (iii) Total load column 1 has to take = $730 \times 10^3 \text{ N}$

A1

A1

- (iv) Total load ground has to take = $(730 + 100) \times 10^3$ M1
 $= 830 \times 10^3 \text{ N}$ A1
- (d) (i) Coordinate (4.3, 922) is treated as anomaly. Best fit line drawn through the rest of the seven points. A1
- (ii) Gradient of line = $\frac{1040 - 916}{5.00 - 4.40}$ M1
 $= 207 \times 10^3 \text{ N m}^{-2}$ A1
- (iii) Factor of safety = $\frac{645 \times 10^3}{207 \times 10^3}$
 $= 3.12$ M1
 Since factor of safety is greater than 2.9, it is safe. A1
- (e) (i) Applied load = $(207 \times 10^3) \pi \left(\frac{2.5}{2}\right)^2$ M1
 $= 1016 \times 10^3 \text{ N}$ A1
- (ii) Total allowable weight of passengers = $(1016 - 830) \times 10^3$
 $= 186 \times 10^3 \text{ N}$ A1
- (iii) Total allowable number of passengers per car = $\left(\frac{186 \times 10^3}{60 \times 10}\right)$ M1
 $= 310$ A1
- (f) Number of passengers a car can take when train arrives at station
 $= 0.25 \times 310 = 77.5$ C1
 Total number of passengers train can take = 77.5×6
 $= 465$ M1
 Longest time interval between train arrivals = $\left(\frac{465}{240}\right)$
 $= 1.94 \text{ minutes}$ A1